

SOcially
Responsible
AI

SRAI 101



Lu Cheng
<https://lcheng.org/>

AI, AI, Everywhere. Why NOW?

- AI is pervasive



Isomorphic Labs to bring AI to the Real World



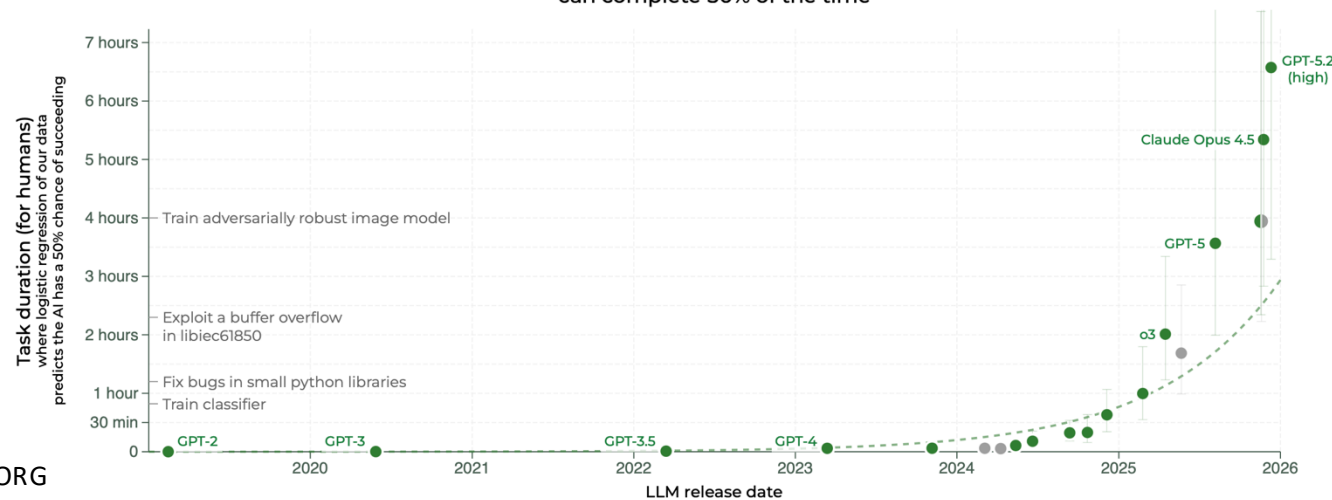
Virtual Assistant, Erica



Robots in CNY Gala

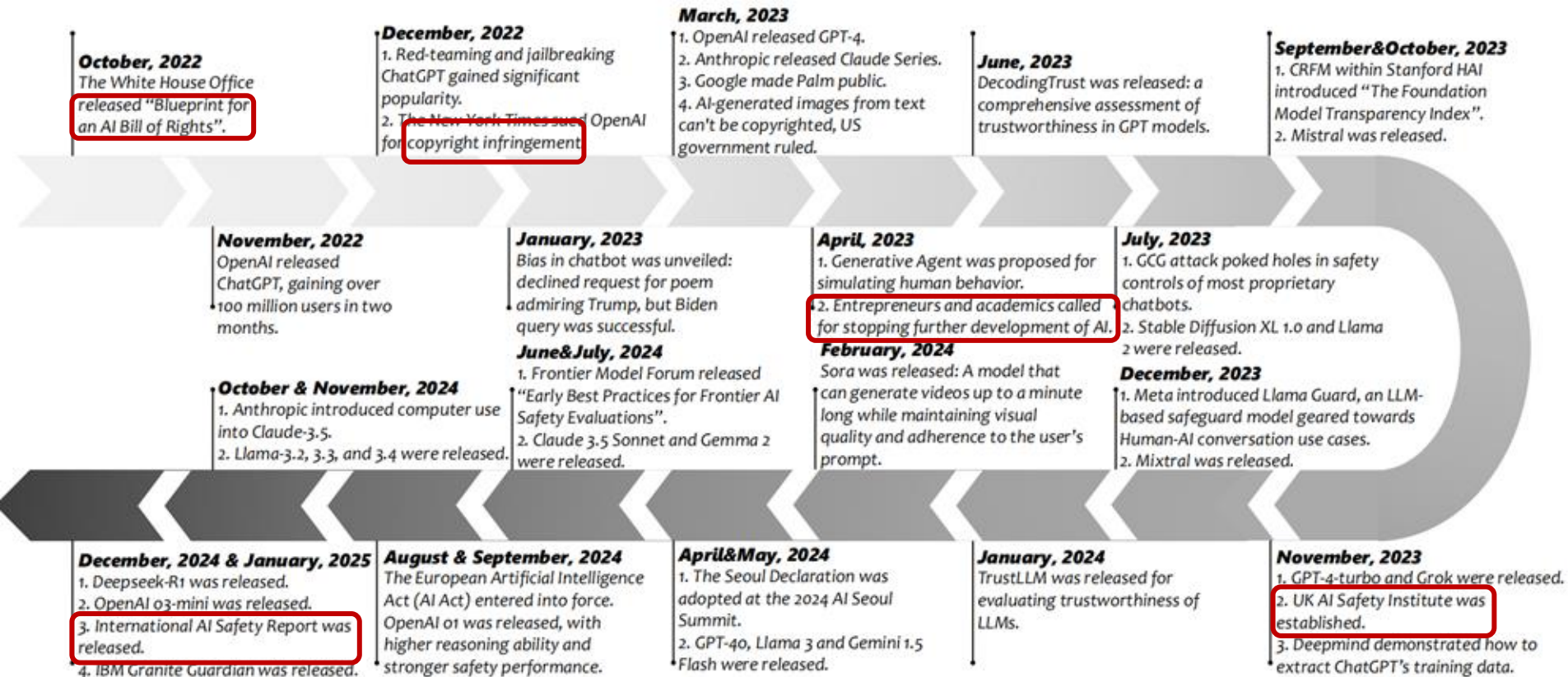
- AI is powerful

The time-horizon of software engineering tasks different LLMs can complete 50% of the time



What is the problem?

Throughout history, the development of models has consistently prompted public scrutiny regarding their **trustworthiness** and **social impact**.





What Could Go WRONG?

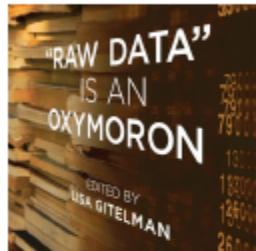
Frame of Reference

Definition: Given data $D = \{x_1, x_2, \dots, x_n\}$, target labels $L = \{y_1, y_2, \dots, y_n\}$ find a hypothesis H such that

$$\operatorname{argmin}_H \sum_{x_i \in D} \ell(H(x_i), y_i)$$

where ℓ is some loss function.

Abstractions



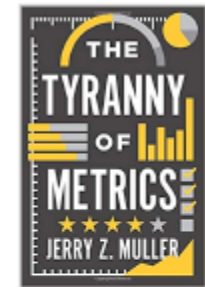
DATA

Transformation of raw data into feature vectors
determines unit of analysis and quantification
Data (especially human data) occurs in context



LABELS

Discretization matters; number and boundaries affect results
Who defines the labels, external "expert", self-report?
Measurement model



LOSS

Interact in non-obvious, domain-specific ways
Penalize errors differently
Often trade-offs are simplified and baked into the loss function

All optimization problems are abstractions, none of them are "correct"

Structural Plausibility

Mapping from features to labels must pass the “smell test”

There should be plausible scientific connection between the input features and the output label

If there is not, **no matter how well your classifier performs, you should reject the hypothesis**

Automated Inference on Criminality using Face Images

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Abstract

We study, for the first time, automated inference on criminality based solely on still face images. Via supervised machine learning, we build four classifiers (logistic regression, KNN, SVM, CNN) using facial images of 1856 real persons controlled for race, gender, age and facial expressions, nearly half of whom were convicted criminals, for discriminating between criminals and non-criminals. All four clas-

people share the belief that the face reveals to reveal innate traits of a person. Aristotle's work Prior Analytics asserted, "It is impossible to infer character from features, if it is granted that body and the mind are changed together by the same affections". Psychologists have known, for as long as a millennium, the human tendency of inferring innate traits and social attributes (e.g., the trustworthiness, dominance) of a person from his/her facial appearance, and a robust consensus of individuals' infer-

Doesn't pass the plausibility test

Responsible Data Science, Lise Getoor, *IEEE Big Data Keynote*, 2019

Big Data

High dimensional data is problematic because danger of overfitting is significantly higher than low dimensional data



Curse of dimensionality: our intuitions break down in high dimensions



Likelihood of finding random subset of features that are predictive is high



Required **sample size** for generalization proportional to dimension



Multiple testing, p-value hacking, reproducibility crisis

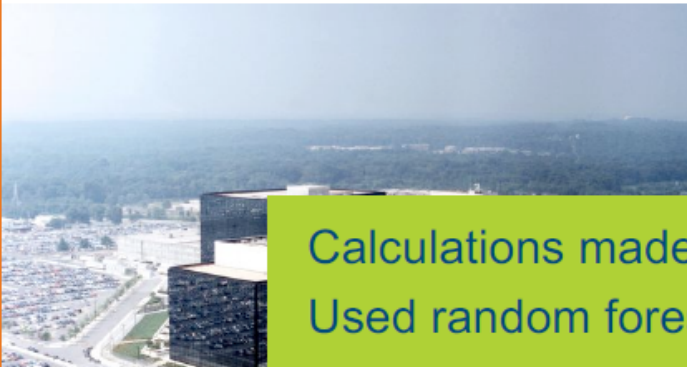
Models learned from high dimensional data treated with extreme care!

MILITARY

Can The NSA's Machines Recognize A Terrorist?

The big problem with little data

By Dave Gershgorn February 16, 2016



Calculations made on 80 variables for each cell phone user
Used random forests

To test their model, project uses data from just 7 known terrorists, plus random sample of 100,000 mobile phone users

When run it the wild, it identified an Al Jazeera reporter covering Al Qaeda as a potential terrorist

MY ALGORITHM HAS AN ACCURACY OF 78.6%**

ACCORDING TO OUT OF SAMPLE TEST WHICH ASSUMES

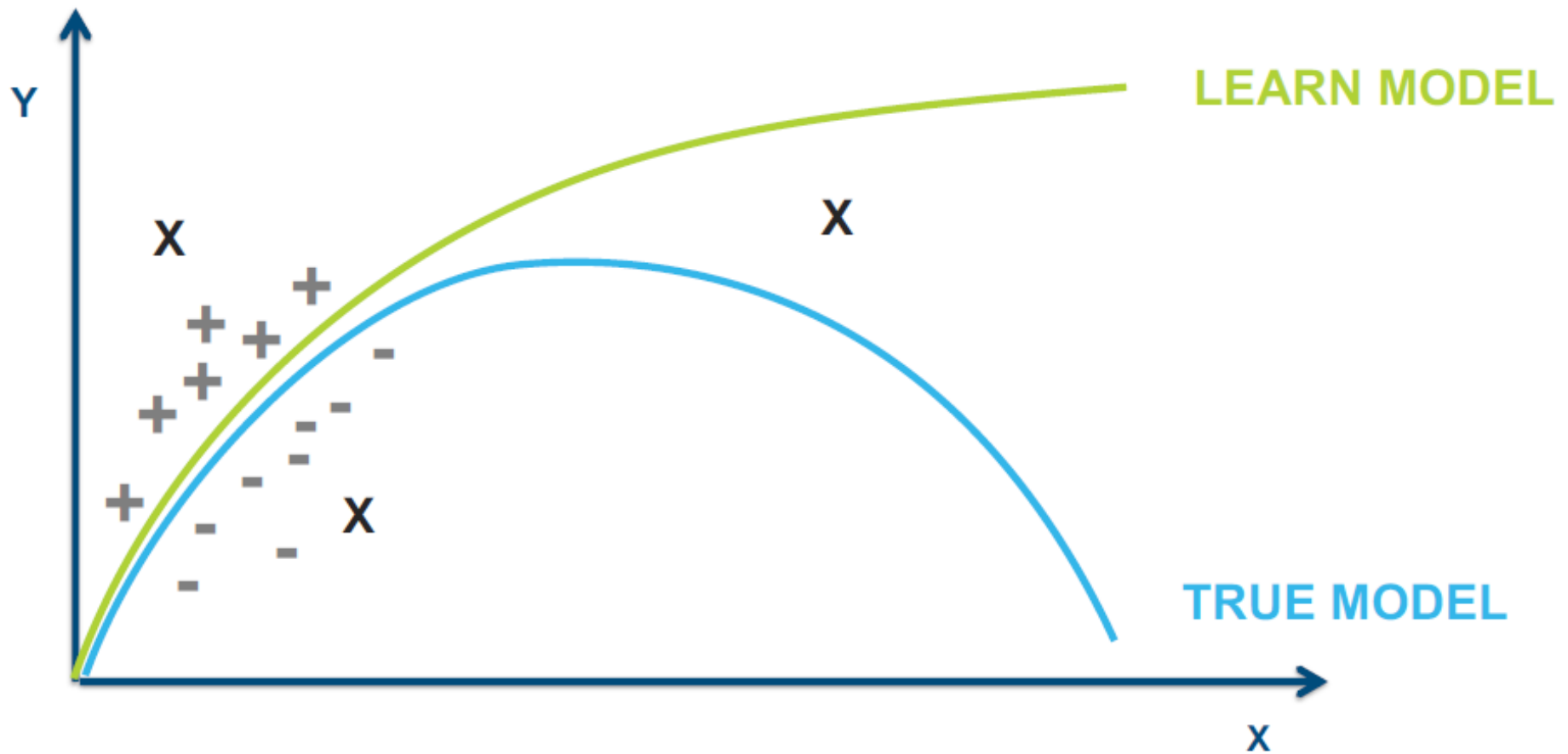
A well-defined population

The training data is a representative sample of the population

The test data is a representative sample of the population

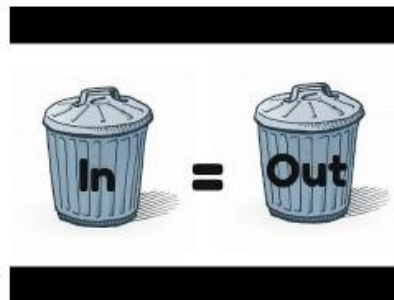
Almost NEVER Hold

NON-REPRESENTATIVE SAMPLES



Data Bias

If the input to the system is biased,
due to selection bias, institutional bias, or societal bias,
then the output will be biased



Famous Computer Science phrase: GARBAGE IN, GARBAGE OUT

Algorithm Bias

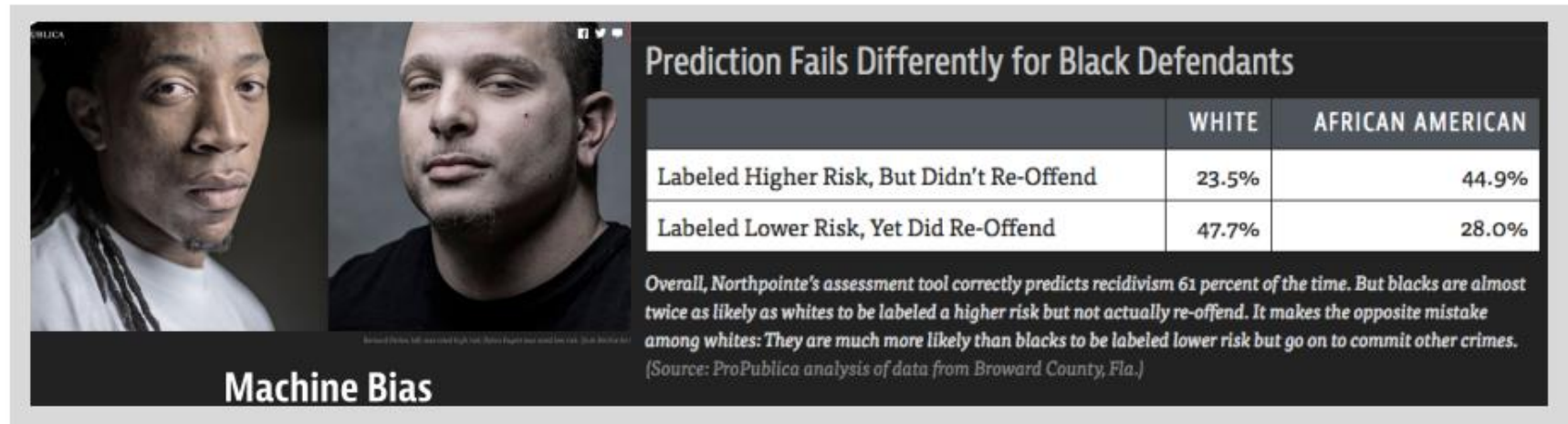
COMPAS

RECIDIVISM RISK

defendant's likelihood of committing a crime

USED THROUGHOUT
CRIMINAL JUSTICE SYSTEM

pretrial, bail and sentencing



Overpredicts recidivism for African Americans; underpredicts recidivism for whites

[ProPublica](#), 2016

Responsible Data Science, Lise Getoor, *IEEE Big Data Keynote*, 2019

Automation Bias

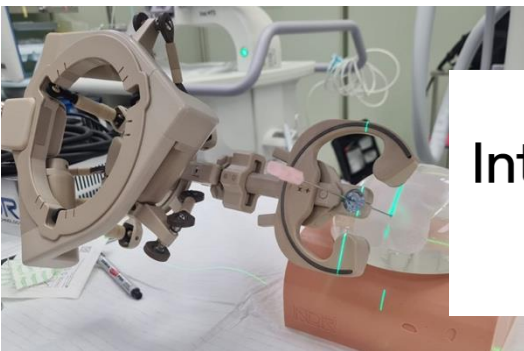
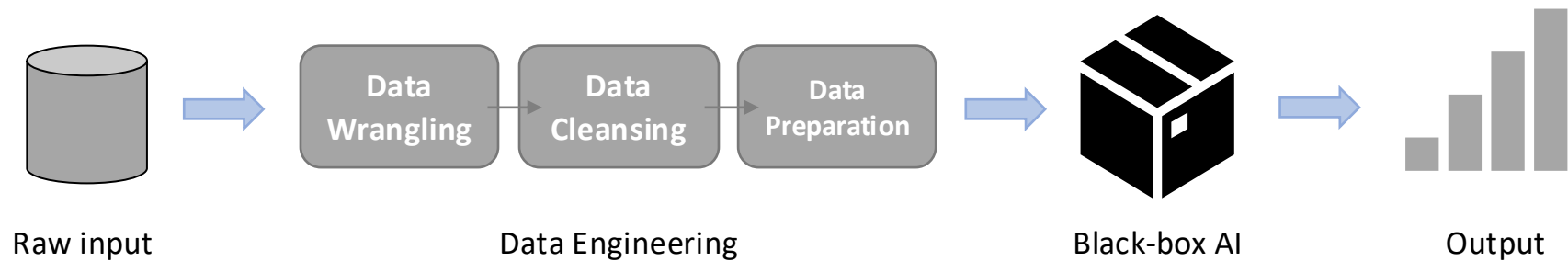
Humans tend to favor suggestions from automated decision-making systems and often ignore contradictory information



- Decision makers may abdicate decision responsibility to algorithms
- Especially for hard decisions, tempting to rely on an algorithm
- This affects risks and accountability



Black-box AI



AI-powered surgical robot

January 7, 2026 Product

Introducing ChatGPT Health

A dedicated experience in ChatGPT designed for health and wellness.

[Join the waitlist ↗](#)

AI-based health chatbot

MEET Jack

- Age: 66
- Diabetic
- Heart attack 6 years ago
- Humana Member

Doctor: "How do you like Care Angel?"

Jack: "I enjoy it. I love it because it calls me every Monday morning with reminders."

"Do you think it's helpful?"

"Yes, I love it! She helps me stay healthy."

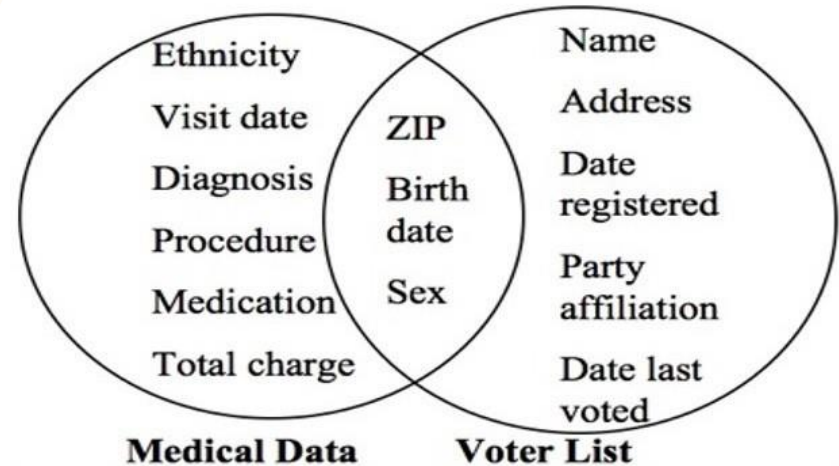
Virtual nursing assistants

AI Can Outperform Doctors. So Why Don't Patients Trust It?

William Weld vs Latanya Sweeney

Massachusetts Group
Insurance Commission
(1997): Released
Anonymized medical
history of state employees

Latanya Sweeney (MIT grad
student): \$20 – Cambridge
voter roll



born July 31, 1945
resident of 02138

64%

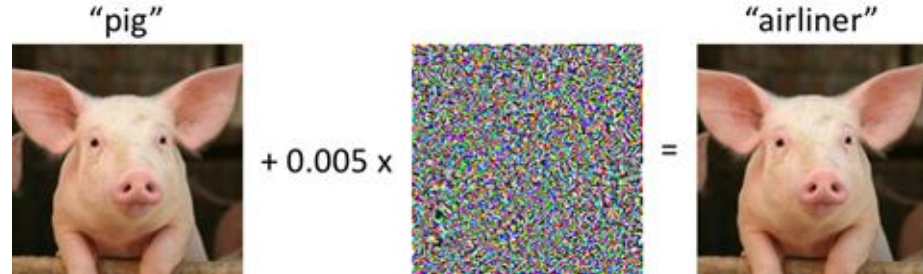
Uniquely identifiable with ZIP
+ birth date + gender (in the
US population)

Golle, "Revisiting the Uniqueness of Simple Demographics in the US Population", WPES 2006

A History of Privacy Failures ...

- Re-identification [Sweeney '00, ...]
 - GIC data, health data, clinical trial data, DNA, Pharmacy data, text data, registry information, ...
- Blatant non-privacy [Dinur, Nissim '03], ...
- Auditors [Kenthapadi, Mishra, Nissim '05]
- AOL Debacle '06
- Genome-Wide association studies (GWAS) [Homer et al. '08]
- Netflix award [Narayanan, Shmatikov '09]
- Social networks [Backstrom, Dwork, Kleinberg '11]
- Genetic research studies [Gymrek, McGuire, Golan, Halperin, Erlich '11]
- Microtargeted advertising [Korolova '11]
- Recommendation Systems [Calandrino, Kiltzer, Naryanan, Felten, Shmatikov '11]
- Israeli CBS [Mukatren, Nissim, Salman, Tromer '14]
- Attack on statistical aggregates [Homer et al. '08] [Dwork, Smith, Steinke, Vadhan '15]

- AI algorithms can be *fragile* under attacks
 - A STOP sign became 45-Mile speed limit



- Ability to understand and control how your data is used
- Ability to get paid for use of your data
- Network effects: People's data can increase in value nonlinearly when combined with other peoples' data

DC AG sues Meta CEO Zuckerberg over Cambridge Analytica privacy scandal

PUBLISHED MON, MAY 23 2022·10:50 AM EDT | UPDATED MON, MAY 23 2022·11:37 AM EDT



Lauren Feiner
@LAUREN_FEINER

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Twitter fined \$150m in US for selling users' data

By Shiona McCallum
Technology reporter

🕒 27 May

Responsible AI Pillars

Fairness



Privacy

Transparency

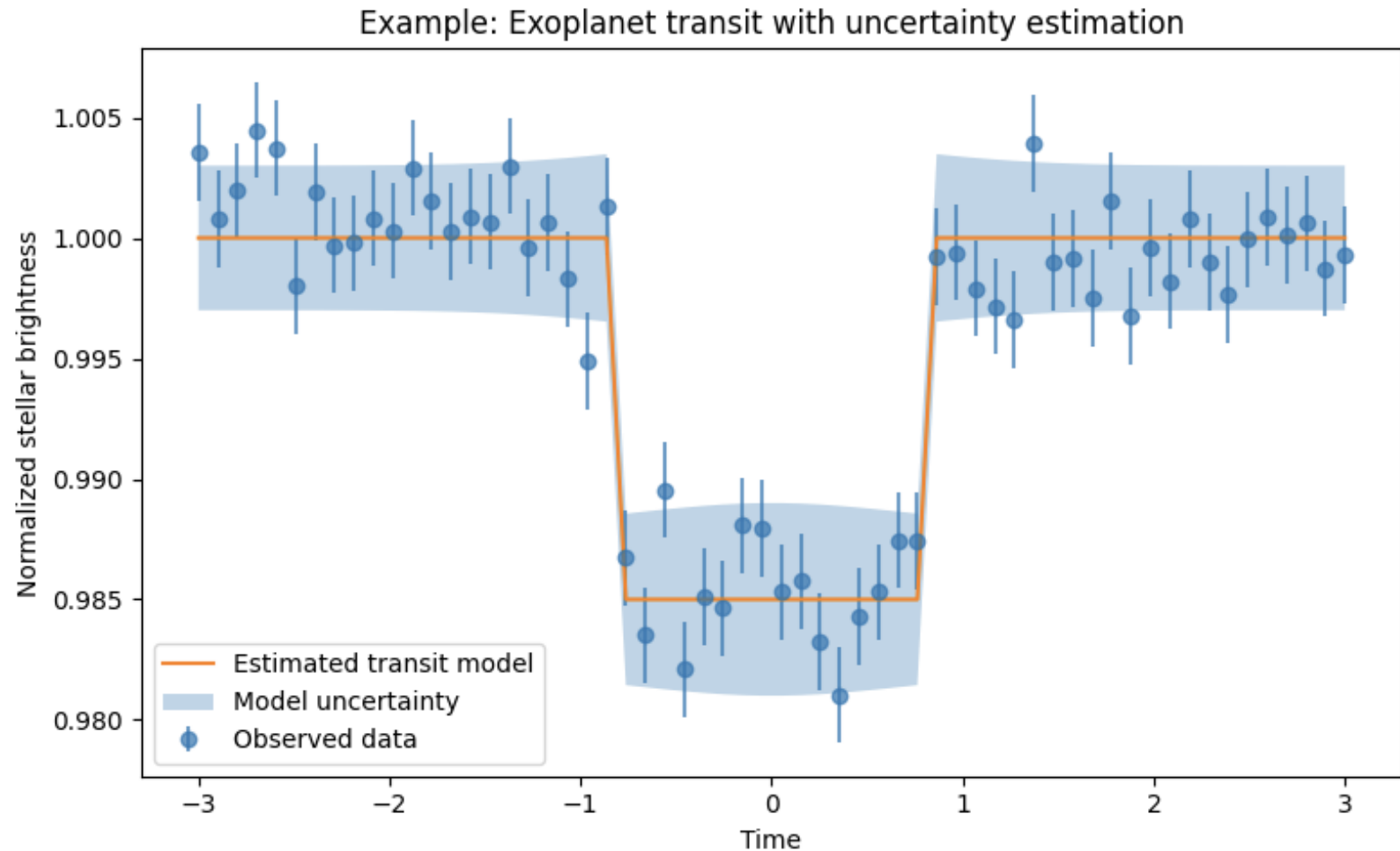


Reliability

Conformal Prediction

- On the importance of UQ
- Set-valued predictions
- Formulation and an example
- Evaluating conformal predictors
- Applications

Why do we need uncertainty estimates in astrophysics?



Earth-like? Super-Earth? Mini-Neptune?

Why do we need uncertainty estimates?

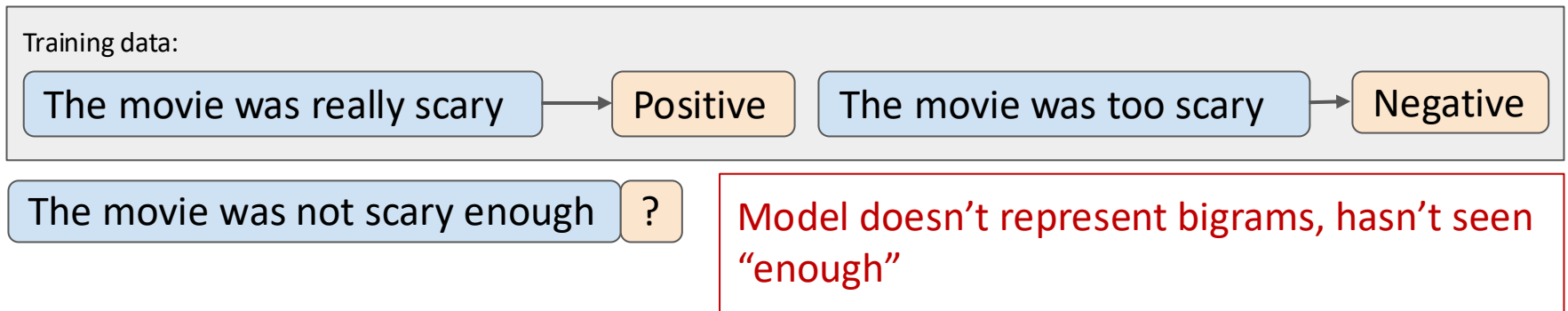
Reliable estimates of **uncertainty** can help us:

-  **Build or reduce trust** in certain pointwise predictions...
-  **Compare** the performance of different models (i.e., uncertainty in metrics)...
-  **Identify areas of improvement** for a given model (e.g., for active learning)...
-  **List all plausible answers** subject to specified probabilistic guarantees...
-  **Produce more natural responses** (that reflect confidence) for dialogue agents...
-  **Abstain** from making predictions when in doubt...
-  ... and more (concrete examples to follow).

Understanding uncertainty sources

Uncertainty sources: Modeling

- Often referred to as **epistemic/model** uncertainty.
- Your model assumptions might be wrong, leading to systematic errors.
- You may lack training data for the type of prediction you are trying to make.



- Two models M_1/M_2 can both perfectly fit the training data but predict differently on x_{test} .
- Epistemic uncertainty can be reduced by collecting more data + improving the model.

Uncertainty sources: Data

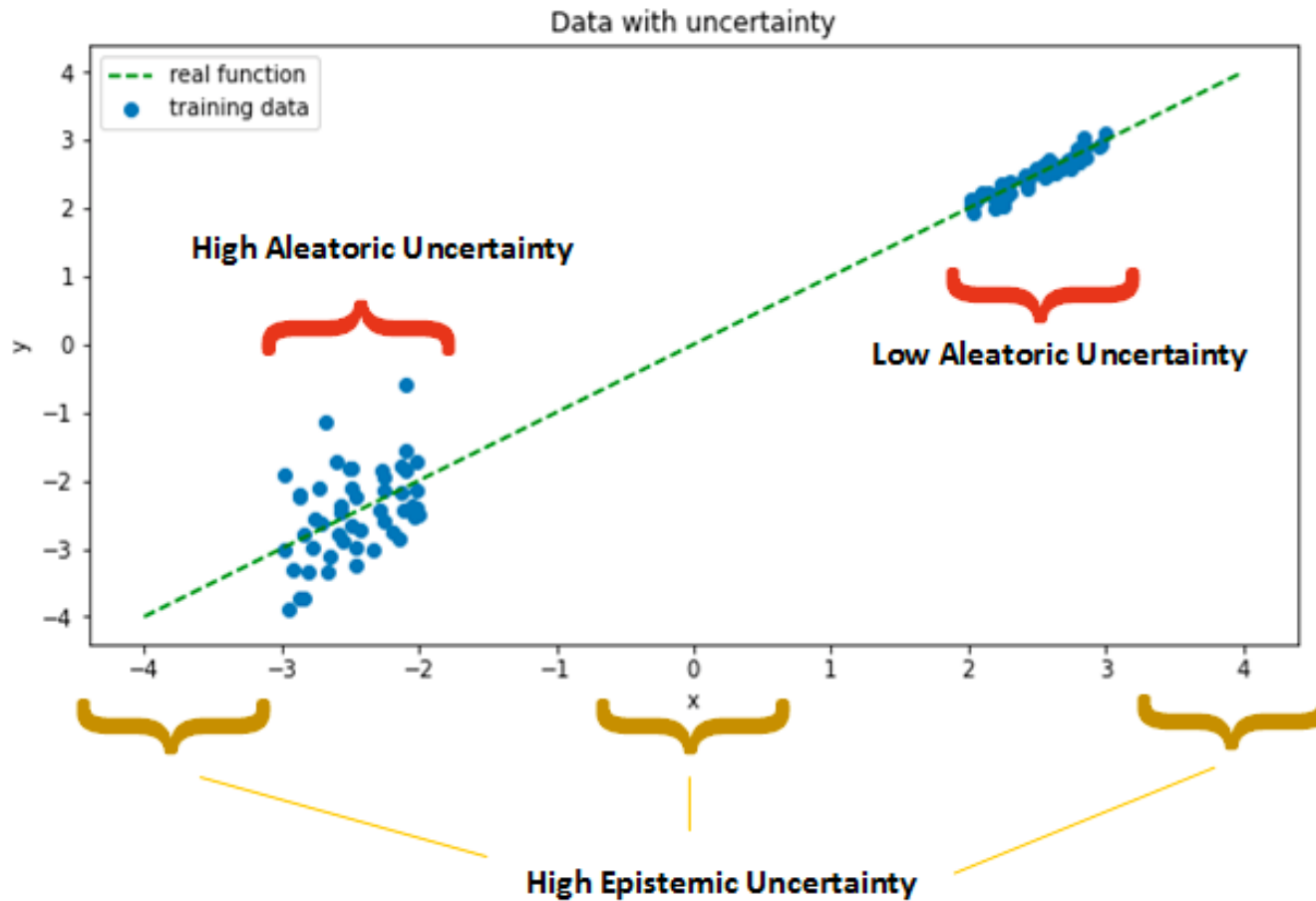
- Often referred to as **aleatoric/data** uncertainty.
- Sometimes certain tasks/data points have an objective/deterministic ground truth:
- But other times, the uncertainty is inherent in the task data itself (stochasticity).

A. Quantum shot noise B. Thermal fluctuations C. Seismic vibrations



Collecting more data or changing the model won't reduce the uncertainty.

Uncertainty sources



[Image by [Michel Kana](#)]

CP: Uncertainty-aware set-valued predictions

- Main idea: instead of calibrating the probabilities of individual outcomes assign calibrated probabilities to **sets of possible outcomes** (e.g., $Y \in \{1, 2, 3\}$).
- Main goal: predict a *small set of plausible answers*, one of which is likely to be correct.
- What's **powerful**: we can calibrate the process for constructing this set such that it provably contains the right answer with **specifiably high probability**.
 - ... in finite samples (i.e., not asymptotically as data $\rightarrow \infty$!).
 - ... for any data distribution (i.e., not assuming particular properties).
 - ... for any underlying model (e.g., doesn't have to be in some family or format).

A motivating example

- Information Retrieval for fact-checking:



(over 6 million possibly relevant English articles in Wikipedia)

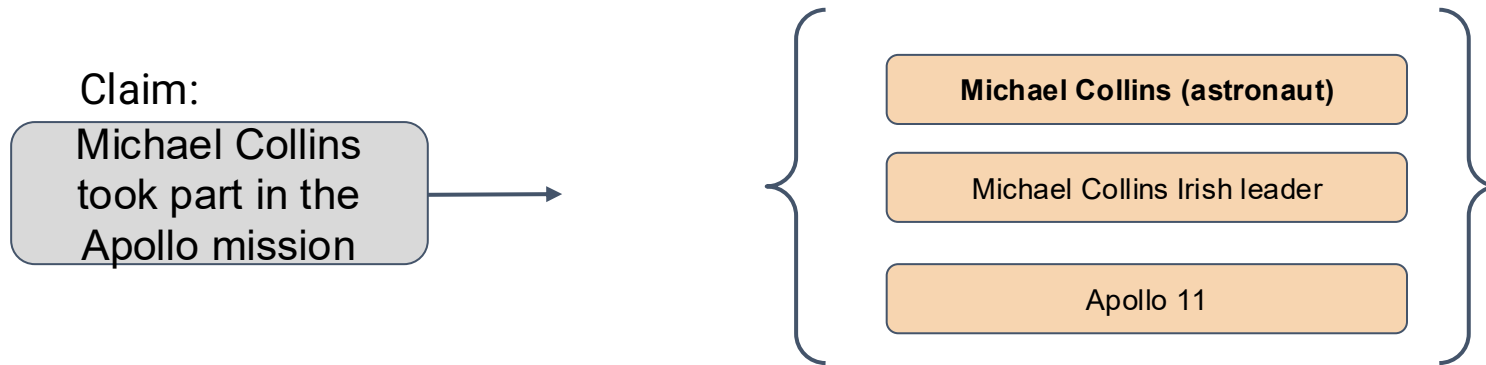
A motivating example

- Information Retrieval for fact-checking



A motivating example

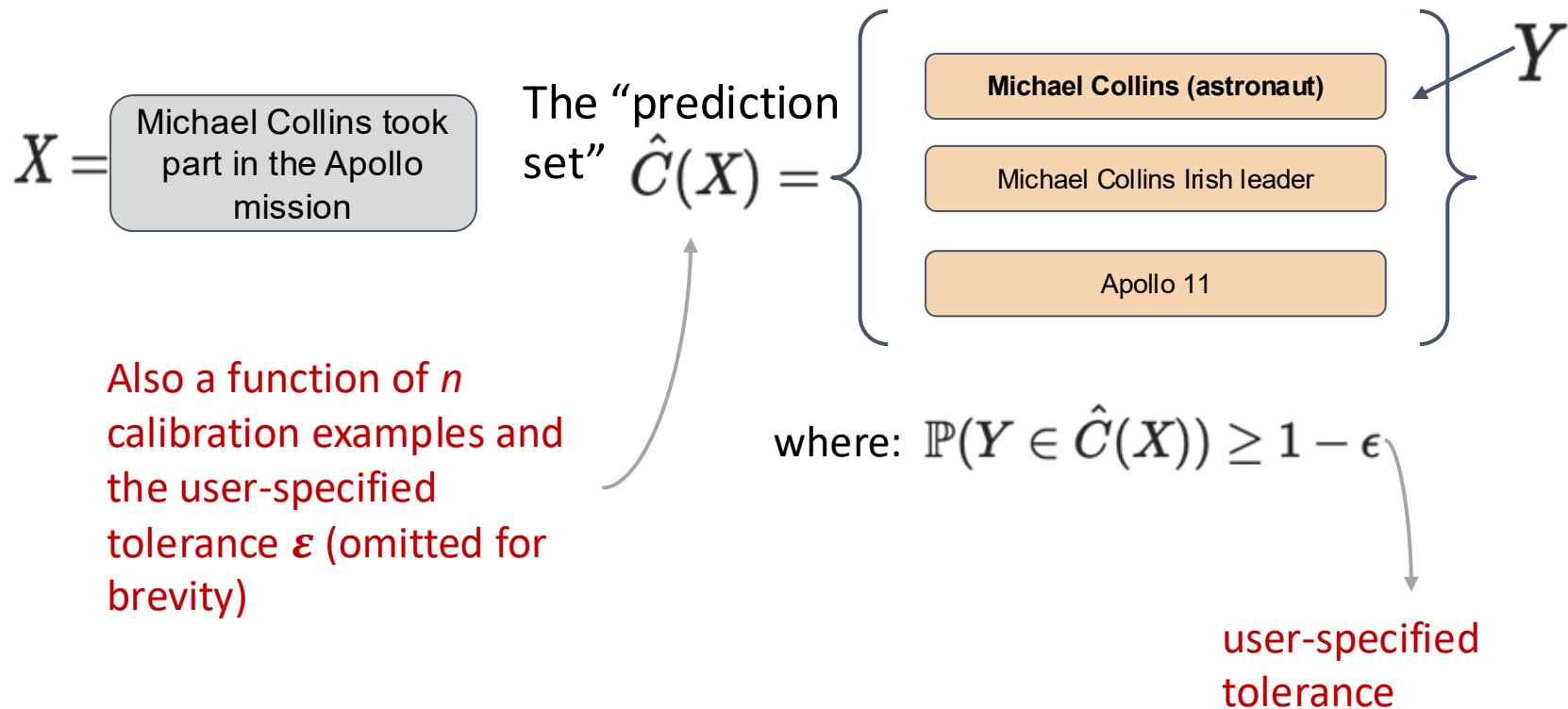
- Information Retrieval for fact-checking



We want (at least) one of these answers to be **correct!**

A motivating example

- More formally, let's introduce some notation:



Conformal prediction framework

- Given n exchangeable examples $(X_i, Y_i) \in \mathcal{X} \times \mathcal{Y}$ and a desired significance level ϵ , for a new input X_{n+1} , return a **set of predictions** $C_n(X_{n+1}) \subseteq \mathcal{Y}$.
- A predictor is **valid** if $C_n(X_{n+1})$ covers the correct label Y_{n+1} w.p. $1 - \epsilon$:

$$\mathbb{P}(Y_{n+1} \in C_n(X_{n+1})) \geq 1 - \epsilon$$

- An **efficient** predictor should satisfy:

$$\mathbb{E}[|C_n(X_{n+1})|] \ll |\mathcal{Y}|$$

Conformal prediction framework

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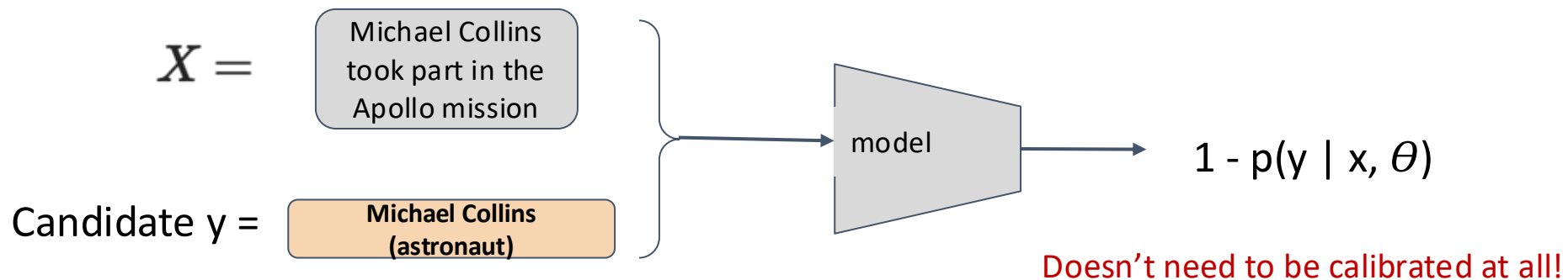
This is
controllable.

How conformal prediction works

- Informally, conformal prediction uses “**nonconformity**” scores to measure surprise.
- Basic idea: suppose I “assign” a candidate label $\mathbf{y}'$ to a given input $\mathbf{x}$. How “strange” does this $(\mathbf{x}, \mathbf{y}')$ pair look relative to other examples that I know to be correct?
- If it is *relatively* strange, it is considered to be **nonconforming** to the dataset.
- If it is *relatively* “not that strange”, then it **conforms** (and we can’t rule $\mathbf{y}'$ out).

Nonconformity measures

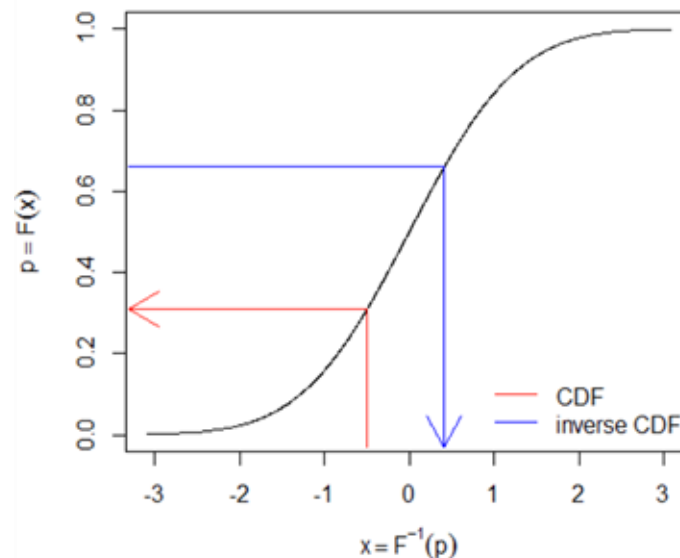
- Can be any mapping $f : (\text{given dataset} \times \text{new example}) \rightarrow \mathbb{R}$.
- Could simply be the loss of an (uncalibrated) classifier trained on the dataset.



- Now, how should we define “relatively” nonconforming?

Review of quantiles and quantile functions

- Quantile functions **invert the cumulative distribution function** of a random variable Z .
- In other words, what value partitions an α -fraction of my dataset examples?
- Example: the 0.5 quantile of the standard normal distribution is 0.



[Image [source](#)]

Review of quantiles and quantile functions

- Formally, for a distribution F_Z , we can write:

$$\text{Quantile}(\alpha; F_Z) := \inf\{z : \mathbb{P}(Z \leq z) \geq \alpha\}$$

- In words, this returns the **smallest value z** such that the probability that a new, randomly drawn $Z \sim F_Z$ is at most z , is **at least α** .
- This gives us a good way of defining “relatively strange”! The probability of randomly drawing a new $Z > z$ from the same distribution by chance is only $1 - \alpha$.

Conformal prediction in action

- Assume we have n “supervised” random data points for calibration: $(X_1, Y_1), \dots, (X_n, Y_n)$.
- We assume that they are **exchangeable** with some distribution (their order doesn’t matter).
- Now, we take note of the distribution of their nonconformity scores V_i ...

Claim 1	Claim 2	Claim 3	Claim 4	Claim 5
Document relevant to claim 1	Document relevant to claim 2	Document relevant to claim 3	Document relevant to claim 4	Document relevant to claim 5
$V_1 = 0.09$	$V_2 = 0.05$	$V_3 = 0.13$	$V_4 = 0.24$	$V_5 = 0.36$

$$\text{Quantile}(0.8; V_{1:5} \cup \{\infty\}) = 0.36$$

Conformal prediction in action

- Test all possible documents for a new, random claim X_{n+1} .
- For each candidate y' , we compute $V_{n+1} = f(X_{n+1}, y')$
- We reject any document with nonconformity

$$V_{n+1} > \text{Quantile}(1 - \epsilon; V_{1:n} \cup \{\infty\})$$

New claim (X_6):

Michael Collins
took part in the
Apollo mission

Candidate
document:

Moon

$$V_6 = 0.73 \quad \times$$

$C(X_6)$ for $\epsilon = 0.2$

$$\text{Quantile}(0.8; V_{1:5} \cup \{\infty\}) = 0.36$$

Conformal prediction in action

- Test all possible documents for a new, random claim X_{n+1} .
- For each candidate y' , we compute $V_{n+1} = f(X_{n+1}, y')$
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$$V_{n+1} > \text{Quantile}(1 - \epsilon; V_{1:n} \cup \{\infty\})$$

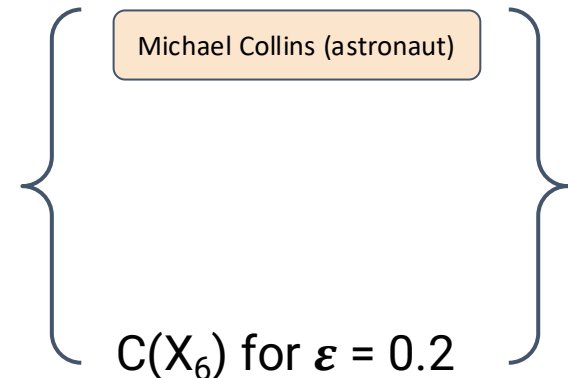
New claim (X_6):

Michael Collins
took part in the
Apollo mission

Candidate
document:

Michael Collins (astronaut)

$$V_6 = 0.14 \quad \checkmark$$



$$\text{Quantile}(0.8; V_{1:5} \cup \{\infty\}) = 0.36$$

Conformal prediction in action

- Test all possible documents for a new, random claim X_{n+1} .
- For each candidate y' , we compute $V_{n+1} = f(X_{n+1}, y')$
- We reject any document with nonconformity

$$V_{n+1} > \text{Quantile}(1 - \epsilon; V_{1:n} \cup \{\infty\})$$

New claim (X_6):

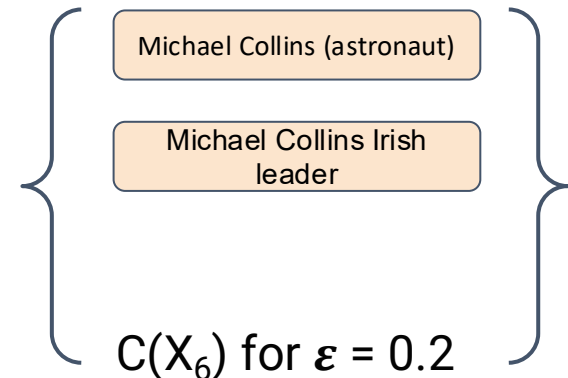
Michael Collins
took part in the
Apollo mission

Candidate
document:

Michael Collins Irish
leader

$$V_6 = 0.34 \quad \checkmark$$

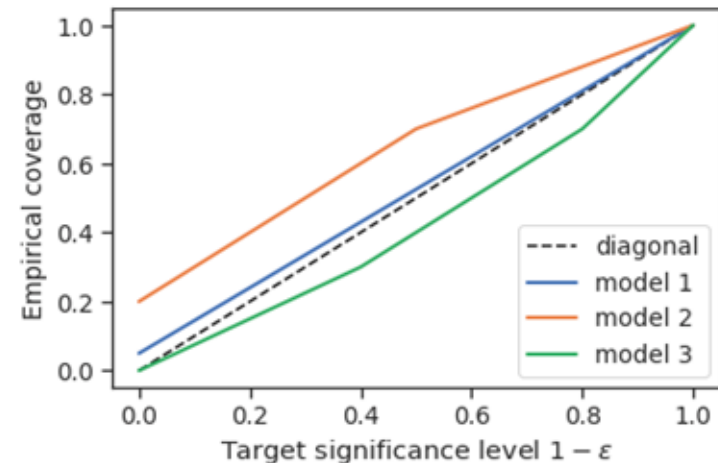
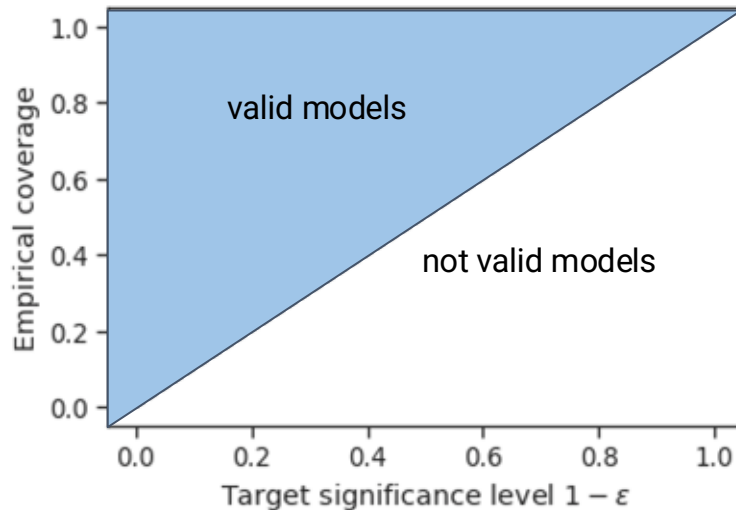
(assuming some imperfect model)



$$\text{Quantile}(0.8; V_{1:5} \cup \{\infty\}) = 0.36$$

Evaluating conformal prediction

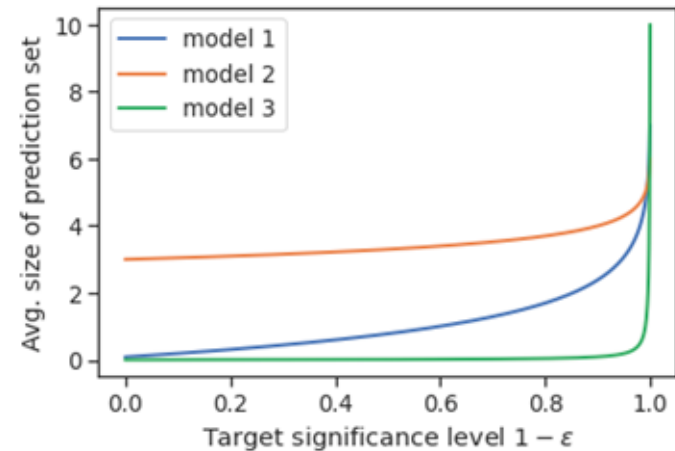
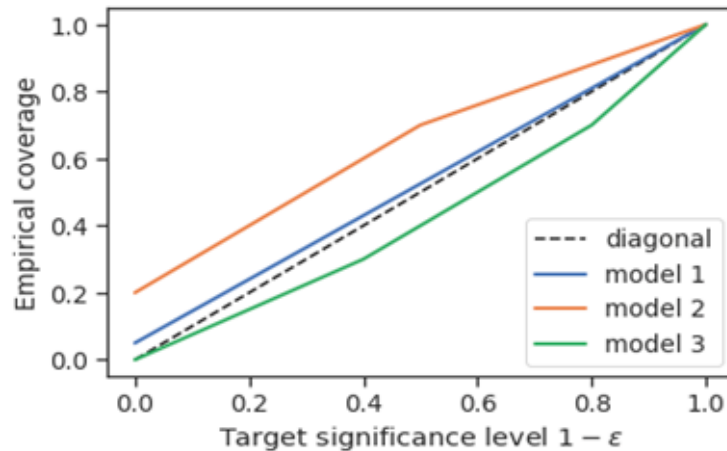
- Empirically verifying the model's **validity** (anything above the diagonal):



- Both models 1 and 2 are valid. Though, model 2 might be too conservative...
- Model 3 is not valid.

Evaluating conformal prediction

- Measuring the **size of prediction sets** (the lower the better):



- While both models 1 and 2 are valid, model 1 is better because the prediction sets are smaller/tighter.
- Model 3 is not valid, therefore, its small prediction sets aren't reliable.

Applications

API Is Enough: Conformal Prediction for Large Language Models Without Logit-Access

EMNLP'24

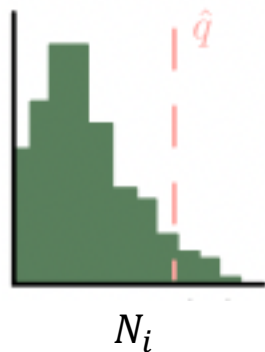
Jiayuan Su, Jing Luo, Hongwei Wang, Lu Cheng



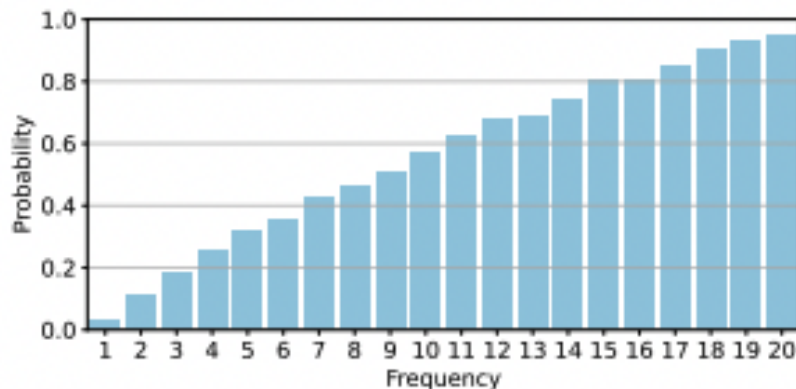
Challenges of Adapting CP to LLMs

- Many LLMs are API-access, no access to predicted probabilities
- Even with predictive probabilities, they are known to be miscalibrated → degraded CP performance

Coarse-grained Uncertainty → Frequency as a Proxy



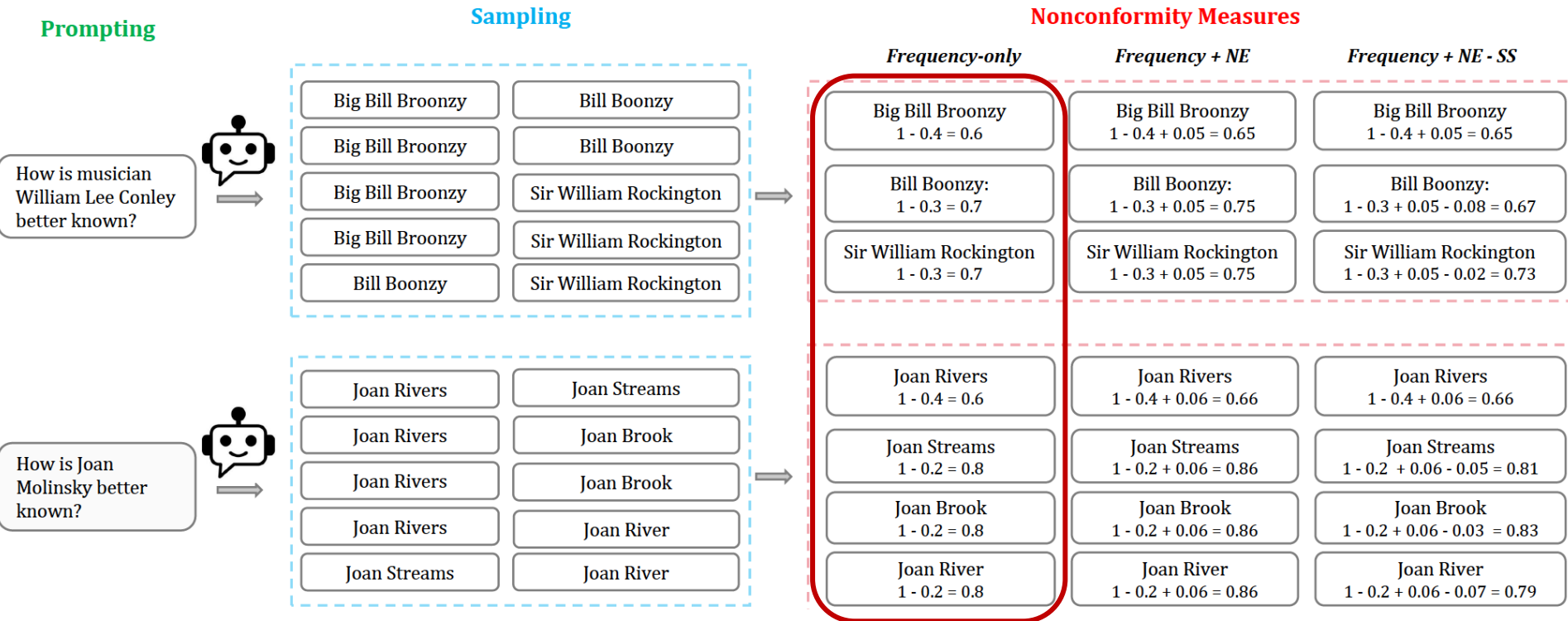
Ranking
Matters!



$$F(\hat{y}_a^{(i)}, m) = \frac{\tilde{p}[\hat{y}_a^{(i)}]}{m}$$

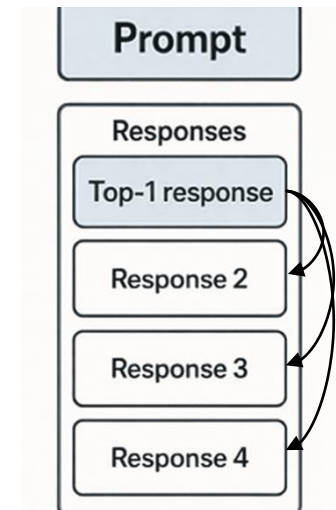
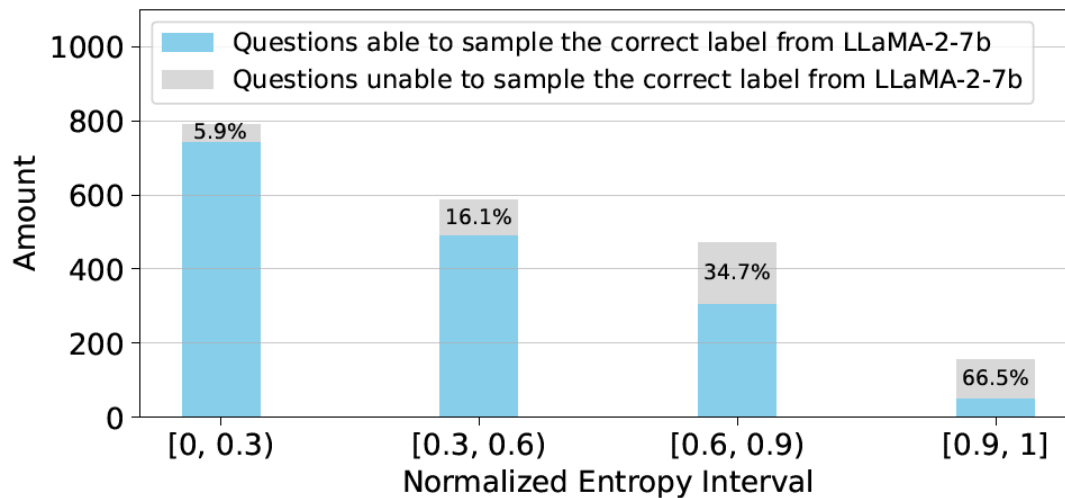
Positive correlation between response frequency and average real probability

Challenges Cont. - The Concentration Issue



Fine-grained Uncertainty

- Normalized Entropy (NE) for **Question**-wise Uncertainty
- Semantic Similarity (SS) for **Response**-wise Uncertainty



$$H(x^{(i)} | \{\hat{y}_j^{(i)}\}_{j=1}^m) = \left| \frac{\sum_{a=1}^n \tilde{F}(\hat{y}_a^{(i)}) \log(\tilde{F}(\hat{y}_a^{(i)}))}{\log m} \right|$$

$$SS(\hat{y}_a^{(i)}, P_{\text{highest}}^{(i)}) = \frac{\mathbf{v}(\hat{y}_a^{(i)}) \cdot \mathbf{v}(P_{\text{highest}}^{(i)})}{\|\mathbf{v}(\hat{y}_a^{(i)})\| \cdot \|\mathbf{v}(P_{\text{highest}}^{(i)})\|}$$

Final nonconformity score:

$$N^{(i)} = -F(\hat{y}_a^{(i)}, m) + \boxed{\lambda_1} \cdot H(x^{(i)} | \{\hat{y}_j^{(i)}\}_{j=1}^m) \\ - \boxed{\lambda_2} \cdot SS(\hat{y}_a^{(i)}, P_{\text{highest}}^{(i)}), \quad \text{Hyperparameters}$$

Proposition 3.2 (Coverage guarantee of LofreeCP).

Suppose $(X_i, Y_i)_{i=1, \dots, n}$ and $(X_{\text{test}}, Y_{\text{test}})$ are i.i.d. Let $C_{1-\alpha}(X_{\text{test}})$ be defined as in Step 3. Then we have the coverage guarantee:

$$P \{Y_{\text{test}} \in C_{1-\alpha}(X_{\text{test}})\} \geq 1 - \alpha,$$

where $\alpha \in (0, 1)$ denotes the desired error rate.

Experiments – A Glimpse

- *Empirical Coverage Rate (ECR)*
 - *Size-Stratified Coverage (SSC,↑)*
 - *Average Prediction Set Size (APSS,↓)*
- Coverage
 - Efficiency (i.e., set size)

Methods	Logit-Access	Error Rate								
		0.2			0.25			0.3		
		ECR	SSC↑	APSS↓	ECR	SSC↑	APSS↓	ECR	SSC↑	APSS↓
First-K _{white}	✓	82.1	76.6	3.39	76.1	72.9	1.90	✗	✗	✗
CLM	✓	80.2	73.4	2.29	75.2	69.1	1.55	70.1	68.3	1.28
SCP	✓	80.3	75.7	2.25	75.1	70.0	1.59	70.3	74.5	1.21
SAPS	✓	80.0	77.9	2.74	75.1	64.2	1.80	70.0	49.4	1.55
First-K _{black}	✗	80.1	76.8	2.70	76.4	72.2	1.90	✗	✗	✗
LofreeCP (Ours)	✗	80.1	79.0	2.19	75.3	74.5	1.43	70.3	76.7	1.08

Methods	Logit-Access	Error Rate								
		0.35			0.4			0.45		
		ECR	SSC↑	APSS↓	ECR	SSC↑	APSS↓	ECR	SSC↑	APSS↓
First-K _{white}	✓	✗	✗	✗	62.4	62.5	1.00	✗	✗	✗
CLM	✓	65.0	69.3	0.96	60.1	72.7	0.81	55.2	83.3	0.70
SCP	✓	65.1	76.4	1.02	60.3	75.7	0.85	55.3	82.5	0.74
SAPS	✓	65.1	57.4	1.28	60.1	70.7	0.85	55.1	76.5	0.72
First-K _{black}	✗	66.5	66.5	1.00	✗	✗	✗	✗	✗	✗
LofreeCP (Ours)	✗	65.1	78.5	0.90	60.0	81.0	0.75	55.2	84.1	0.66

Differentiable Conformal Training for LLM Reasoning Factuality

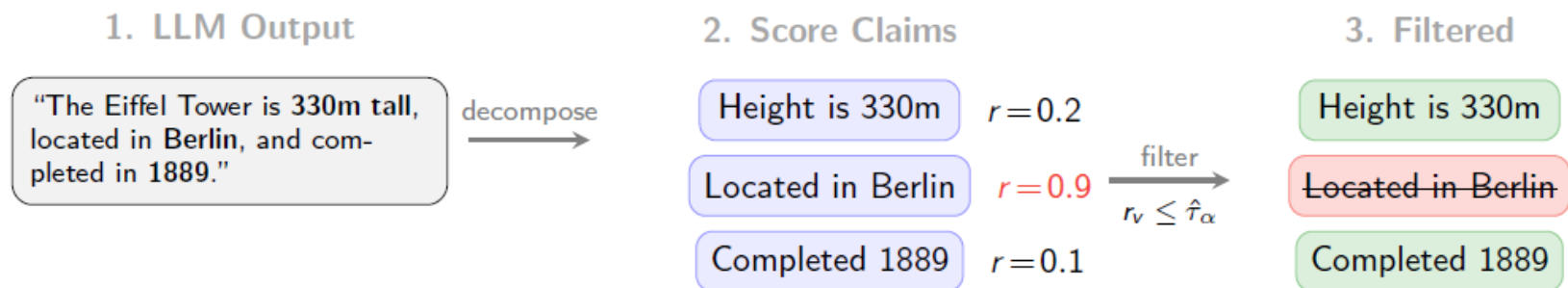
ICML'26

Nathan Hittesdorf, Marco Salzetta, Lu Cheng



Conformal Prediction for LLM Reasoning Factuality

Idea: Decompose LLM output into claims, score each, filter risky ones



Risk score r_v : e.g., self-consistency
(Low frequency across generations $\rightarrow$ high risk)

Threshold $\hat{\tau}_\alpha$: calibrated via CP
Keep claim iff $r_v \leq \hat{\tau}_\alpha$

Guarantee

With probability $\geq 1 - \alpha$, retained claims are factually correct

The Problem: Reasoning Claims Are Not Independent

- Independent filtering fails for multi-step reasoning!
- **Solution:** Represent LLM outputs as **Approximate Deducibility Graphs (ADGs)**

Problem: The endpoints of a segment are (1, 4) and (1, 10). What is the sum of the coordinates of the midpoint?

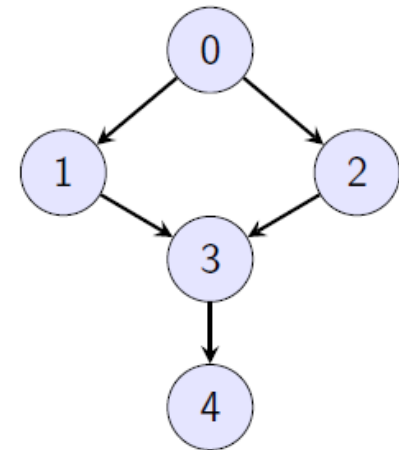
0: Midpoint formula: $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$

1: x-coordinate: $\frac{1+1}{2} = 1$

Claims: **2:** y-coordinate: $\frac{4+10}{2} = 7$

3: Midpoint is (1, 7)

4: Sum: $1 + 7 = 8$

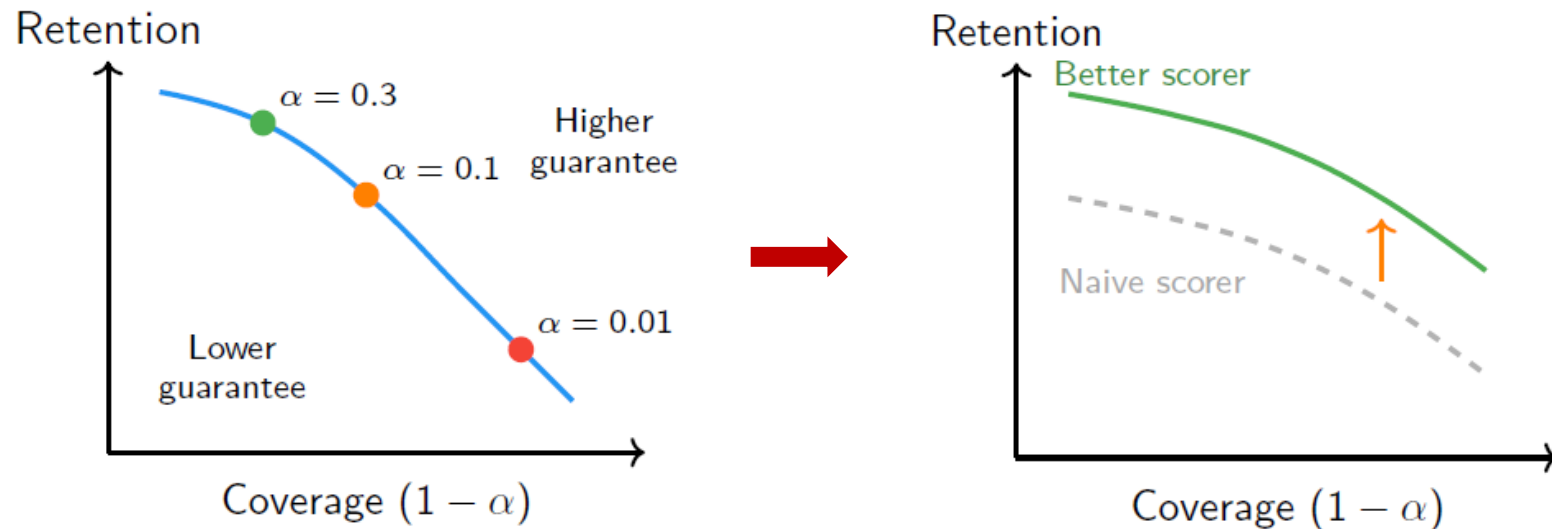


Nodes = claims, Edges = dependencies

- **Key idea:** Find the **largest threshold** τ where all subgraphs are coherently factual

The Retention-Guarantee Tradeoff

Problem: Stronger guarantees $\Rightarrow$ fewer claims retained



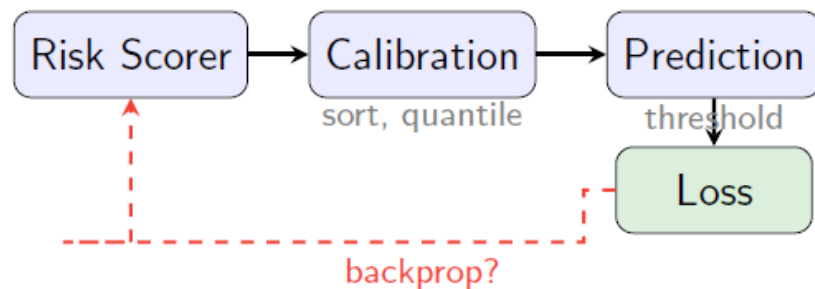
Key insight: Better risk scores $\Rightarrow$ better tradeoff curve!

Our Goal

Train a risk scorer to **maximize claim retention** while maintaining coverage guarantees

Conformal Training: The Challenge

Goal: Train a risk scorer by backpropagating through conformal prediction

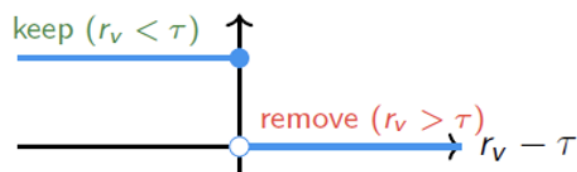


Optimize θ to maximize retention

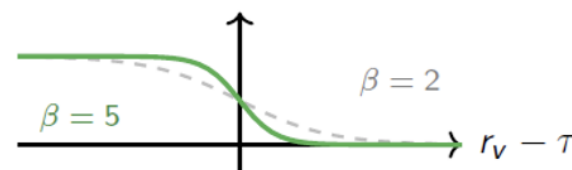
Solution: Differentiable Relaxations

Replace discrete operations with smooth approximations that preserve semantics

Hard Indicator $\mathbb{I}[r_v \leq \tau]$

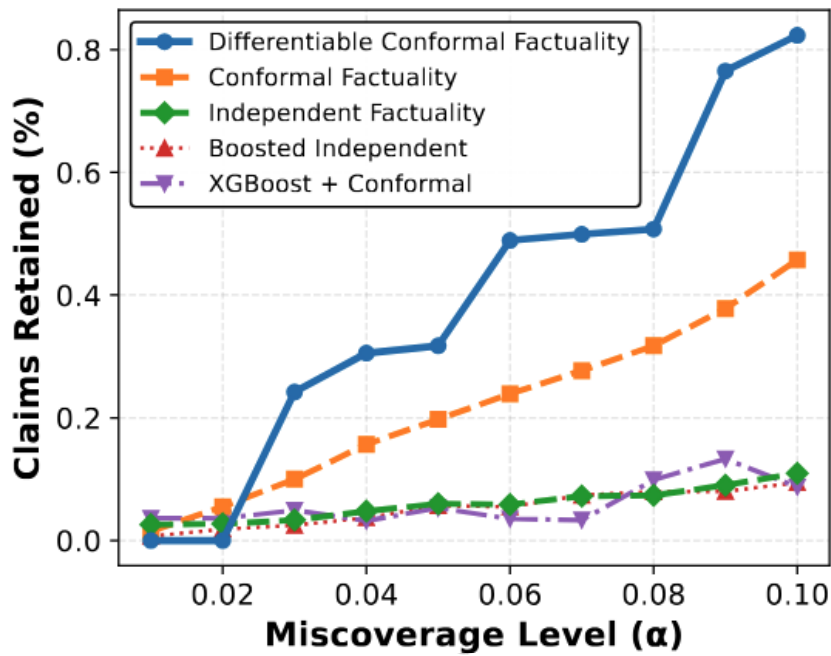


Soft Sigmoid $\sigma(\beta(\tau - r_v))$



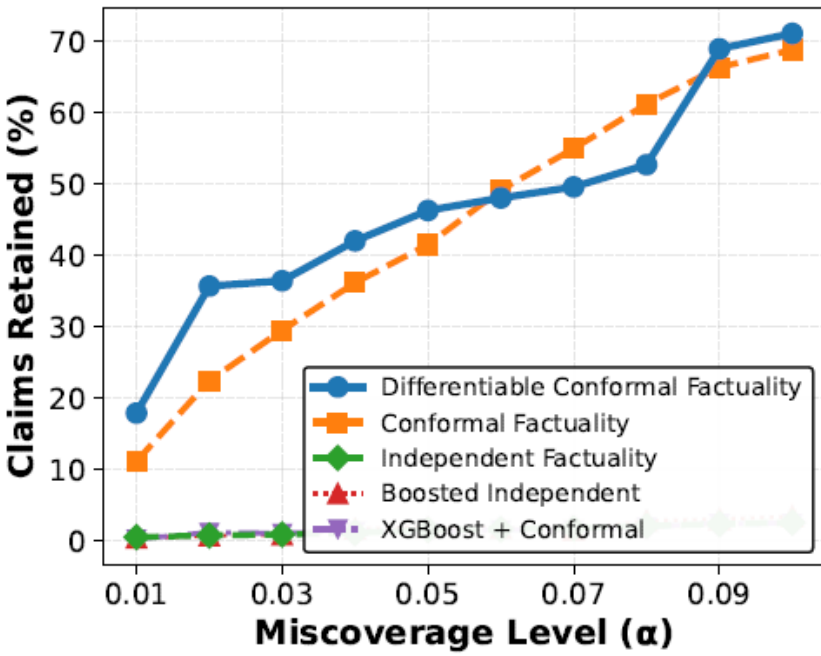
Results: Improved Retention with Coverage Guarantees

MATH Dataset



+141% retention vs. frequency baseline

FELM Dataset



+61% retention vs. frequency baseline

Key Findings

DCF consistently **maintains coverage guarantees** while substantially improving claim retention



Conformal Prediction: A Data Perspective

ACM CSUR'25

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 YU GUI, The University of Chicago, Chicago, United States
 LU CHENG, Computer Science, University of Illinois Chicago, Chicago, United States

♥ CP-Router: An Uncertainty-Aware Router Between LLM and LRM

AAAI'26 (Oral)

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NeurIPS'24

Conformalized Link Prediction on Graph Neural Networks

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Conformalized Time Series with Semantic Features

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NeurIPS'23

Robust Uncertainty Quantification for Self-Evolving Large Language Models via Continual Domain Pretraining

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Equal Opportunity of Coverage in Fair Regression

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Temporal Graph Prototype-conditioned Conformal Prediction for Fraud Detection

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- https://colab.research.google.com/github/GitHubLuCheng/CP-tutorial/blob/main/TP_cp_init_student_version.ipynb
- https://colab.research.google.com/github/GitHubLuCheng/CP-tutorial/blob/main/TP_cp_classif_init_student_version.ipynb



Questions

- Can you think of an example where AI has **positively** or **negatively** impacted society, and what factors do you think contributed to that outcome?
- What **responsibilities** do you think developers or companies had in this situation?
- How might this outcome have been **avoided** or **improved**?